

NutriCoach Service Contracts

API Specifications & Service Boundaries

1. Calc Engine Service

Type: Internal Library (No HTTP endpoints)

Location: lib/calc-engine.ts

Interface

```
// Types
interface ProfileInput {
  age: number; // 13-120
  sex: 'male' | 'female';
  weightKg: number; // 20-500
  heightCm: number; // 50-300
  activityLevel: 'sedentary' | 'light' | 'moderate' | 'active' | 'very_active';
  goal: 'lose' | 'maintain' | 'gain';
  macros?: MacroDistribution;
}

interface MacroDistribution {
  carbs: number; // 0-100, must sum to 100
  protein: number; // 0-100
  fat: number; // 0-100
}

interface NutritionResult {
  bmr: number; // Basal Metabolic Rate
  tdee: number; // Total Daily Energy Expenditure
  targetCalories: number; // Goal-adjusted calories
  macros: {
    carbsG: number;
    proteinG: number;
    fatG: number;
  };
  formulaVersion: string; // For audit trail
}

interface ValidationResult {
  valid: boolean;
  errors: string[];
}

// Functions
function calculateNutrition(input: ProfileInput): NutritionResult;
function adjustCaloriesForGoal(tdee: number, goal: Goal): number;
function calculateMacroGrams(calories: number, dist: MacroDistribution): MacroGrams;
function validateProfileInput(input: unknown): ValidationResult;
```

Constraints

- **Pure Functions Only:** No database access, no side effects
 - **Deterministic:** Same input always produces same output
 - **Versioned:** Formula changes tracked via `formulaVersion`
 - **AI Cannot Modify:** AI Coach can READ results but CANNOT change formulas
-

2. Plan Service

Base Path: `/api/meal-plan`

Endpoints

Generate Meal Plan

```
POST /api/meal-plan/generate
Authorization: Bearer {session token}
Content-Type: application/json

{
  "calculationId": "calc abc123", // Optional, uses latest if omitted
  "preferences": {
    "excludeIngredients": ["peanuts", "shellfish"],
    "cuisinePreferences": ["mexican", "mediterranean"],
    "mealCount": 3,
    "includeSnacks": true
  }
}
```

Response (202 Accepted - Async):

```
{
  "jobId": "job_xyz789",
  "status": "queued",
  "estimatedTime": 30,
  "pollUrl": "/api/meal-plan/job/job_xyz789"
}
```

Response (200 OK - Sync for simple plans):

```
{
  "id": "plan_abc123",
  "days": [
    {
      "day": 1,
      "meals": [
        {
          "type": "breakfast",
          "name": "Protein Oatmeal",
          "foods": [...],
          "nutrition": { "calories": 450, "protein": 30, ... }
        }
      ]
    }
  ],
  "shoppingList": [...],
  "totalNutrition": { ... }
}
```

Get Meal Plan

```
GET /api/meal-plan/{id}
Authorization: Bearer {session token}
```

Get Shopping List

```
GET /api/meal-plan/{id}/shopping
Authorization: Bearer {session token}
```

3. Challenge/Streak Service

Base Path: /api/challenges , /api/streak

Challenge Endpoints

List Challenges

```
GET /api/challenges
Authorization: Bearer {session token}

Query Parameters:
- status: active | upcoming | completed
- limit: number (default: 10)
- offset: number (default: 0)
```

Response:

```
{
  "challenges": [
    {
      "id": "chal_7day_protein",
      "name": "7-Day High Protein Challenge",
      "description": "Hit your protein goals for 7 days straight",
      "durationDays": 7,
      "requirements": {
        "dailyProteinGoal": 120,
        "minMealsLogged": 3
      },
      "rewards": {
        "points": 500,
        "badge": "protein_champion"
      },
      "userStatus": "not_joined" || "active" || "completed"
    }
  ],
  "total": 15
}
```

Join Challenge

```
POST /api/challenges/{id}/join
Authorization: Bearer {session token}
```

Get Progress

```
GET /api/challenges/{id}/progress
Authorization: Bearer {session token}
```

Response:

```
{
  "challengeId": "chal_7day_protein",
  "userId": "user_abc",
  "startedAt": "2026-03-01T00:00:00Z",
  "currentDay": 3,
  "dailyProgress": [
    { "day": 1, "completed": true, "proteinG": 135, "mealsLogged": 4 },
    { "day": 2, "completed": true, "proteinG": 128, "mealsLogged": 3 },
    { "day": 3, "completed": false, "proteinG": 45, "mealsLogged": 1 }
  ],
  "overallProgress": 0.43,
  "onTrack": true
}
```

Streak Endpoints

Get Current Streak

```
GET /api/streak
Authorization: Bearer {session token}
```

Response:

```
{
  "currentStreak": 12,
  "longestStreak": 45,
  "lastActivityDate": "2026-03-03",
  "todayCompleted": true,
  "activitiesRequired": [
    { "type": "LOG_MEAL", "completed": true },
    { "type": "AI_INTERACTION", "completed": true }
  ]
}
```

Record Activity

```
POST /api/streak/activity
Authorization: Bearer {session token}
Content-Type: application/json

{
  "activityType": "LOG MEAL" | "AI INTERACTION" | "PLAN VIEW" | "CHECK IN"
}
```

4. Leaderboard Service

Base Path: /api/leaderboard

Endpoints

Get Weekly Leaderboard

```
GET /api/leaderboard/weekly
Authorization: Bearer {session_token} (optional for public)

Query Parameters:
- limit: number (default: 50, max: 100)
- offset: number (default: 0)
```

Response:

```
{
  "periodKey": "2026-W09",
  "lastUpdated": "2026-03-03T12:00:00Z",
  "rankings": [
    {
      "rank": 1,
      "userId": "user_abc",
      "displayName": "FitJohn",
      "avatarUrl": "https://i.ytimg.com/vi/LBIr1RXiko4/sddefault.jpg",
      "points": 2450,
      "stats": {
        "streakDays": 15,
        "mealsLogged": 42,
        "challengesCompleted": 2
      }
    }
  ],
  "userRank": {
    "rank": 156,
    "points": 890
  },
  "total": 12500
}
```

Get All-Time Leaderboard

```
GET /api/leaderboard/all-time
```

Get Challenge Leaderboard

```
GET /api/leaderboard/challenge/{challengeId}
```

Get User Rank

```
GET /api/leaderboard/user/{userId}/rank
```

Response:

```
{
  "userId": "user_abc",
  "weeklyRank": 156,
  "weeklyTotal": 12500,
  "allTimeRank": 892,
  "allTimeTotal": 45000,
  "percentile": 98.2
}
```

5. Share Service

Base Path: /api/share

Endpoints

Create Share Link

```
POST /api/share/challenge/{challengeId}
Authorization: Bearer {session token}
Content-Type: application/json

{
  "includeStats": true,
  "customMessage": "Join me on this challenge!"
}
```

Response:

```
{
  "shareId": "share_abc123",
  "slug": "fit-john-7day-x3k9",
  "url": "https://nutricoach.app/c/fit-john-7day-x3k9",
  "ogImageUrl": "https://images.fittfinder.com/668ee3cb-868d6d07338498388e06570168c79fb44ba0df3f0ba2259662b8b30/original.jpg",
  "expiresAt": "2026-04-03T00:00:00Z"
}
```

Get Public Payload (SSR)

```
GET /api/share/{slug}
```

Response:

```
{
  "type": "challenge",
  "title": "FitJohn's 7-Day Protein Challenge",
  "description": "Day 5 of 7 - 87% progress!",
  "user": {
    "displayName": "FitJohn",
    "avatarUrl": "..."
  },
  "challenge": {
    "name": "7-Day High Protein Challenge",
    "progress": 0.87,
    "currentDay": 5
  },
  "ogImage": "https://i.etsystatic.com/57327234/r/il/e69f50/7314940205/il_1080xN.7314940205_3mzn.jpg",
  "cta": {
    "text": "Join the Challenge",
    "url": "/get-started?ref=fit-john-7day-x3k9"
  }
}
```

Generate Share Card (Async)

```
POST /api/share/generate-card
Authorization: Bearer {session token}
Content-Type: application/json

{
  "shareId": "share abc123",
  "template": "challenge_progress"
}
```

Response:

```
{
  "jobId": "job_card_xyz",
  "status": "queued",
  "pollUrl": "/api/share/card/job_card_xyz"
}
```

6. AI Coach Service

Base Path: /api/chat

Endpoints

Send Message

```
POST /api/chat
Authorization: Bearer {session token}
Content-Type: application/json

{
  "sessionId": "session abc", // Optional, creates new if omitted
  "message": "What should I eat for dinner to hit my protein goal?"
}
```

Response (Streaming):

```
HTTP/1.1 200 OK
Content-Type: text/event-stream

data: {"type":"start","sessionId":"session_abc"}
data: {"type":"chunk","content":"Based on your"}
data: {"type":"chunk","content":" remaining protein goal"}
data: {"type":"tool_call","tool":"get_nutrition_target","result":{"proteinRemaining":45}}
data: {"type":"chunk","content":" of 45g, I recommend..."}
data: {"type":"done","usage":{"promptTokens":150,"completionTokens":89}}
```

Tool Calling Interface

The AI Coach can call these internal tools:


```

interface AICoachTools {
    // Read-only access to user data
    get_user_profile(): UserProfile;
    get_nutrition_target(): NutritionTarget;
    get_today_progress(): DailyProgress;
    get_meal_suggestions(criteria: MealCriteria): MealSuggestion[];

    // Read-only access to food database
    search_foods(query: string): FoodItem[];
    get_food_nutrition(foodId: string): FoodNutrition;

    // Read-only access to challenges
    get_active_challenges(): Challenge[];
    get_challenge_progress(challengeId: string): ChallengeProgress;

    // CANNOT modify calculations, only read
    get_calculation_result(calculationId?: string): NutritionResult;
}

```

Constraint: AI Coach CANNOT:

- Modify nutrition calculations
- Change user profile data
- Log meals on behalf of user
- Join/leave challenges

7. Analytics / Event Collector

Base Path: /api/analytics , /api/acquisition

Event Collection

Track Event (Batched)

```

POST /api/analytics/event
Content-Type: application/json

{
  "events": [
    {
      "type": "PAGE VIEW",
      "data": { "page": "/tools/calorie-calculator" },
      "timestamp": "2026-03-03T10:30:00Z"
    },
    {
      "type": "CALCULATOR USE",
      "data": { "calculatorType": "calorie", "result": { "tdee": 2200 } },
      "timestamp": "2026-03-03T10:31:00Z"
    }
  ],
  "sessionId": "sess anonymous xyz"
}

```

Acquisition Tracking

Track Acquisition

```
POST /api/acquisition/track
Content-Type: application/json

{
  "sessionId": "sess xyz",
  "source": "seo" | "social" | "referral" | "direct" | "product hunt",
  "referrer": "https://google.com/...",
  "landingPage": "/tools/calorie-calculator",
  "utmParams": {
    "source": "google",
    "medium": "organic",
    "campaign": null
  },
  "referralCode": "FITJOHN123" // Optional
}
```

Analytics Dashboard

Get Acquisition Analytics

```
GET /api/analytics/acquisition
Authorization: Bearer {admin token}

Query Parameters:
- startDate: ISO date
- endDate: ISO date
- groupBy: day | week | month
```

Response:

```
{
  "period": {
    "start": "2026-02-01",
    "end": "2026-03-03"
  },
  "summary": {
    "totalVisitors": 45000,
    "totalSignups": 2800,
    "conversionRate": 0.062,
    "bySource": {
      "seo": { "visitors": 25000, "signups": 1500, "rate": 0.06 },
      "social": { "visitors": 8000, "signups": 600, "rate": 0.075 },
      "referral": { "visitors": 5000, "signups": 500, "rate": 0.10 },
      "direct": { "visitors": 5000, "signups": 150, "rate": 0.03 },
      "product_hunt": { "visitors": 2000, "signups": 50, "rate": 0.025 }
    }
  },
  "timeSeries": [
    { "date": "2026-02-01", "visitors": 1500, "signups": 85 },
    ...
  ]
}
```

Error Response Format

All endpoints return errors in this format:

```
{
  "error": {
    "code": "VALIDATION_ERROR" | "AUTH_ERROR" | "NOT_FOUND" | "RATE_LIMITED" | "SERVER_ERROR",
    "message": "Human-readable error message",
    "details": { // Optional
      "field": "age",
      "reason": "Must be between 13 and 120"
    },
    "requestId": "req_abc123" // For debugging
  }
}
```

HTTP Status Codes

Code	Usage
200	Success
201	Created
202	Accepted (async job queued)
400	Validation error
401	Not authenticated
403	Not authorized
404	Resource not found
429	Rate limited
500	Server error

Rate Limits

Endpoint Category	Limit	Window
Auth endpoints	10	1 minute
Chat/AI	30	1 minute
Plan generation	5	1 hour
Calculator APIs	60	1 minute
General APIs	100	1 minute
Analytics (write)	1000	1 minute

Rate limit headers:

```
X-RateLimit-Limit: 100
X-RateLimit-Remaining: 95
X-RateLimit-Reset: 1709467200
```